← Back to Unit 2 Index

QUESTION 6 • UNIT 2 – HYPERPARAMETERS

Why are Hyperparameters used in Neural Networks? What are the methods for tuning hyperparameters?

📝 5 Marks

Hyperparameters in Neural Networks

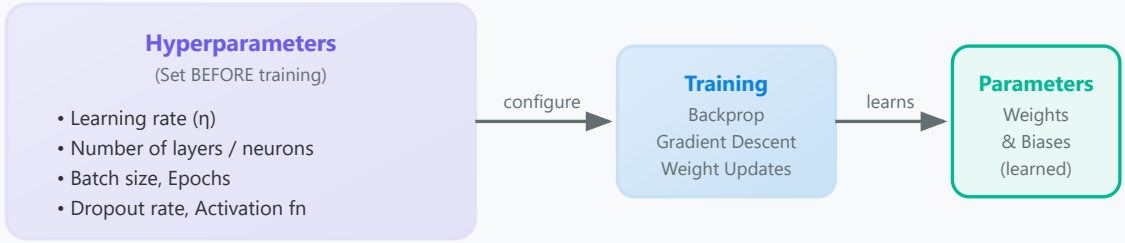
Purpose, types, and tuning methods — with diagrams

What are Hyperparameters?

Hyperparameters are configuration settings for a neural network that are **set before training begins** — unlike model parameters (weights & biases) which are learned during training. They control the learning process and define the model's architecture.

Proper tuning is essential: wrong hyperparameters cause **underfitting** (model too simple) or **overfitting** (model memorises training data).

Figure 1 — Parameters vs. Hyperparameters



Hyperparameters configure the training process; parameters (weights & biases) are learned by it.

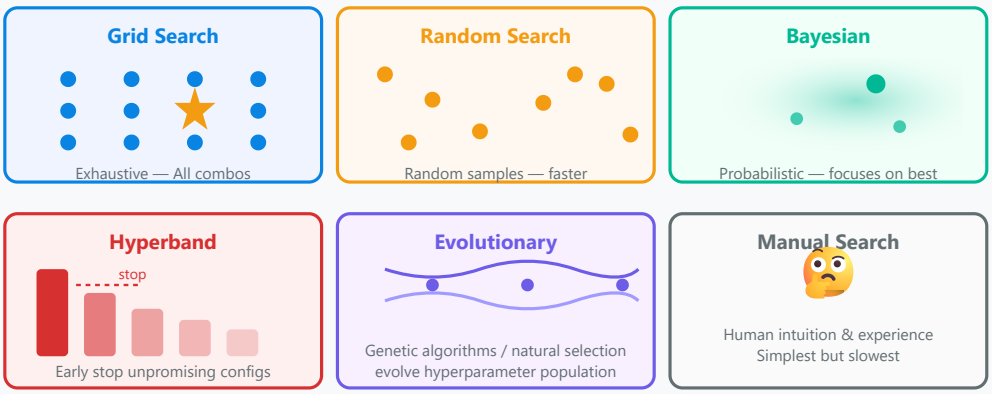
Why Are Hyperparameters Used?

Figure 2 — Five Purposes of Hyperparameters

Hyperparameters serve five key roles from controlling learning speed to ensuring the model generalises well.

Methods for Tuning Hyperparameters

Figure 3 — Hyperparameter Tuning Method Comparison



Six tuning strategies — from manual trial-and-error to intelligent Bayesian optimisation and resource-efficient Hyperband.

1. Manual Search

User sets values based on intuition and experience.

✓ Easy, no extra tools

✗ Time-consuming, may miss optimum

2. Grid Search

Tests all combinations within a predefined range.

✓ Exhaustive — guaranteed to find best in range

✗ Computationally expensive; infeasible for large spaces

3. Random Search

Randomly samples within the hyperparameter range.

✓ More efficient than grid for high-dimensional spaces

✗ May miss optimum due to randomness

4. Bayesian Optimization

Uses probabilistic models to focus on promising regions.

✓ Efficient; focuses search intelligently

✗ Complex to implement; expensive overhead

5. Gradient-Based Optimization

Treats hyperparameters as continuous variables; uses gradients.

✓ Fast for differentiable hyperparameters

✗ Requires differentiability; local minima risk

6. Evolutionary Algorithms

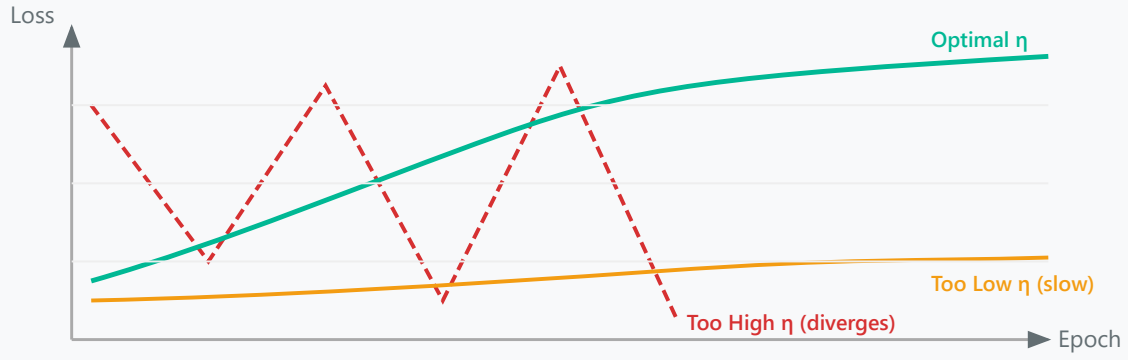
Mimics natural selection to evolve optimal configurations.

✓ Handles non-differentiable, complex spaces

✗ Expensive; the algorithm itself needs tuning

Key Hyperparameters in Neural Networks

Figure 4 — Effect of Learning Rate on Training Loss



Learning rate η is the most critical hyperparameter — too high causes divergence, too low causes very slow convergence, optimal gives smooth fast descent.

Common hyperparameters and their roles:

Hyperparameter	Role	Category
Learning Rate (η)	Controls step size of weight updates	Learning
Batch Size	Samples per forward/backward pass	Learning
Epochs	How many times full dataset is processed	Learning
Number of Layers	Depth of the network	Architecture
Neurons per Layer	Capacity of each layer	Architecture
Activation Function	Non-linearity type used per neuron	Architecture
Dropout Rate	Random neuron deactivation to prevent overfitting	Regularization
L1/L2 Coefficient	Penalises large weights	Regularization